Correcting AIFS Tropical Cyclone Intensity Forecasts: Reproduction, Tail-Aware Learning, and a Probabilistic Extension

Michel Faloughi
SCMP 5990 Independent Study, University of Pennsylvania

October 2026 (draft)

Abstract

AI weather models such as ECMWF's AIFS-Single forecast tropical cyclone tracks well but systematically underestimate intensity, because a 0.25° grid cannot resolve the eyewall. Allen et al. (2026) showed that a simple machine-learned correction, trained on tabular features of the forecast itself, removes most of this bias. This report reproduces the gradient-boosted half of that work at reduced scale, using only public data: 399 operational AIFS-Single forecasts covering every North Atlantic and East Pacific storm from February 2025 to September 2026, tracked with a purpose-built MSLP-minimum tracker and corrected with gradient-boosted trees under storm-grouped cross-validation. The raw model's maximum-wind mean absolute error (MAE) of 30.3 kt is more than halved to 14.4 kt, against 9.4 kt for the official NHC forecast. Raw-model error and the rapid-intensification failure mode match the paper almost exactly (71.4 kt vs. their 71.1 kt raw MAE on RI cases), while the corrected error remains above the paper's 11.0 kt, consistent with training on one and a half seasons instead of nine years. The tracker is validated against ECMWF's own operational AIFS tracks to within 1.1 kt MAE. The second half of the report asks whether a learned correction can be made to respect the observed tail of the intensity distribution, and what that buys. Eta-learning (Chang and Sapsis, 2026), a tail-matching penalty on a small neural network, beats a tuned MSE network by 1.1 kt overall (95% storm-bootstrap CI $[-2.3, -0.03]$) and by 5.6 kt on rapid intensification (RI) cases, and beats the classical post-hoc quantile-mapping baseline by 2.1 kt; most of its advantage over an untuned network, however, is regularization. A diagnosis of *which* storms receive the extreme forecasts shows the penalty controls how many extreme forecasts are issued, not where they go, a limitation shared by any point forecast forced to match the observed marginal. The constructive response is a probabilistic correction: quantile gradient boosting with a cross-conformal recalibration step gives 80% intervals that cover 82% of out-of-storm cases and 84% of a future season, and a median that is 0.8 kt better than the point GBM in cross-validation. Adding SHIPS-type environmental predictors computed from the AIFS pressure-level fields helps by about 1 kt at short leads and not at all on RI, and the derived probability of RI has no skill over climatology with 15 RI storms of training data.

1 Introduction

Data-driven weather models have reached operational skill for synoptic-scale prediction, and ECMWF's AIFS-Single (Lang et al., 2024) runs operationally alongside the physics-based IFS. For tropical cyclones these models inherit a structural weakness: with grid spacing of about 28 km, the inner core of a hurricane is unresolved, so forecast 10 m winds saturate far below observed intensity even when the track is accurate. Allen et al. (2026) ("AIFS-TC") demonstrated that this bias is largely *predictable*: a gradient-boosted model over simple features of the AIFS forecast and the

operationally known initial intensity brings AIFS intensity forecasts to parity with the operational frontier.

This project asks how much of that result survives a drastic reduction in resources: public data only, one and a half hurricane seasons instead of nine years of hindcasts, a home-made tracker, and a laptop. Beyond the replication itself, the by-product is an end-to-end open pipeline (catalog, fetch, track, correct, evaluate) of roughly 900 lines of Python.

The reproduction exposes the weakness that motivates the rest of the report: the corrected forecasts never reach the observed extremes. Trees cannot extrapolate beyond their training targets, so the GBM never forecasts above about 125 kt while nine storms in the sample reached 120 kt or more. Sections 6 to 8 study three responses in turn: a tail-matching training objective (eta-learning), richer inputs (environmental predictors of the kind used by the operational SHIPS scheme), and a probabilistic output with calibrated intervals. Every comparison is reported as a paired difference with a storm-level bootstrap interval, because the effective sample is the number of storms (52), not the number of cases (9149).

2 Data

AIFS forecasts. The ECMWF Open Data AWS mirror (`s3://ecmwf-forecasts`, CC-BY-4.0) retains every operational AIFS-Single run since 2025-02-25. Storm lifetimes from the IBTrACS best-track archive (Knapp et al., 2010) define 399 forecast initializations (00/12 UTC) covering all North Atlantic (NA) and East Pacific (EP) storms of at least tropical-storm strength through 2026-09-08. For each run, only the GRIB messages for mean sea-level pressure and 10 m winds are downloaded via HTTP byte ranges, cropped to 0–60°N, 180°W–0°, at leads 0–168 h every 6 h (7.1 GB total, versus hundreds of GB for the full archive).

Reduction caveats. The stored arrays are a lossy reduction: three surface variables, one basin pair, leads capped at 168 h (the mirror holds 360 h), and winds stored in half precision (error well below 0.1 kt, negligible against forecast error).

Pressure-level fields for environmental predictors. For Section 7, ten further fields were pulled for every run and lead from the same archive (via its Google Cloud mirror, which does not rate-limit byte-range requests): u and v at 850 and 200 hPa, temperature at 200, 500 and 700 hPa, specific humidity at 500 and 700 hPa, and skin temperature. Cropped to the same box this adds 23 GB. The tabular dataset used from Section 6 onward is the full set of 9149 labelled (storm, initialization, lead) cases from 52 storms, which is larger than the homogeneous sample of Section 4 because it does not require an OFCL forecast to exist at the same lead.

Operational archive, not hindcasts. Unlike the paper's controlled hindcast training set, these are the forecasts actually issued in real time. Two consequences matter. First, the model version changes mid-dataset: AIFS-Single 1.0 was operational for the 2025 season and 2.0 (released 2026-05-12) for 2026; a version flag is passed to the correction model and results are reported by season. Second, each run starts from the real-time operational analysis with its data-assimilation cutoff, not from a reanalysis.

Verification and benchmarks. Ground truth is the IBTrACS/ATCF best track (USA agency values). The benchmark is OFCL, the official human forecast of the National Hurricane Center, parsed from the ATCF a-deck archive, which also supplies GDMN, the ensemble mean of Google

DeepMind's cyclone model (Google DeepMind, 2025), and EAIO, ECMWF's own tracker applied to the same AIFS runs.

3 Methods

Tracker. Each forecast is seeded at the best-track position at initialization. At each 6 h step the storm is located at the MSLP minimum within 450 km of a first guess (previous position advected with previous motion) on a lightly smoothed field; intensity is that minimum pressure and the maximum 10 m wind within 250 km. Tracking stops when the low is no longer distinct. Against ECMWF's operational AIFS tracks (EAIO) at matched initialization and lead ($n = 1209$, 24 storms), the tracker agrees to a mean difference of +0.8 kt and an MAE of 1.1 kt, so tracker choice is not a meaningful error source.

Correction model. Following the GBM branch of AIFS-TC, gradient-boosted trees (scikit-learn `HistGradientBoostingRegressor`, 5-seed average) predict the residual between best-track and AIFS maximum wind. The 22 features include the AIFS forecast state and its trends and running extremes, translation speed, basin, day of year, the model-version flag, storm age, the observed intensity and 12/24 h intensity trend at initialization, and the initial AIFS bias. Observed rapid-intensification cases (RI, ≥ 30 kt observed gain in 24 h) are weighted 2× in training, as in the paper. Evaluation uses storm-grouped 5-fold cross-validation, so a storm is never corrected by a model that saw any part of it. A persistence-of-initial-bias baseline (carrying the initialization-time AIFS bias forward unchanged) is included as a sanity check; the GBM beats it at every lead beyond 24 h.

4 Results of the reproduction

Table 1 and Figure 1 give the headline result on the homogeneous sample (cases where raw, corrected, and OFCL forecasts all exist and verify against best track): the correction cuts overall MAE from 30.3 to 14.4 kt, with non-overlapping 95% confidence intervals from a storm-level bootstrap. It does not reach OFCL (9.4 kt) overall, but approaches it at long leads (16.4 vs. 14.1 kt at 120 h). On a separate homogeneous sample containing GDMN ($n = 2022$, 41 storms), DeepMind's dedicated cyclone model scores 9.9 kt, on par with OFCL (9.6), with corrected AIFS at 14.6.

Table 2 breaks the pooled result down. The correction helps in both basins and both seasons. Notably, it performs better on the 2026 season (AIFS-Single 2.0, 12.1 kt) than on 2025 (v1.0, 15.5 kt) despite 2026 contributing less training data, suggesting the newer model's errors are more systematic and therefore easier to learn. Rapid intensification remains the hard case: raw AIFS misses RI by 71.4 kt on average, the correction recovers roughly half of that (38.3 kt), and the human forecast (20.2 kt) stays well ahead.

Figure 2 shows the raw bias at its most dramatic: Hurricane Lowell (EP, 2026) reached 140 kt while every raw AIFS forecast saturated near 60 kt. Figure 3 shows the underlying fields and tracks for one initialization with three simultaneous storms.

Table 1: Maximum-wind MAE (kt) versus best track by forecast lead time, homogeneous sample ($n = 2223$ cases, 46 storms, NA+EP, Feb 2025–Sep 2026). Bottom row: all leads pooled, with 95% storm-bootstrap confidence intervals.

Lead (h)	n	AIFS raw	AIFS+GBM	NHC OFCL
12	442	26.4	10.8	4.9
24	396	28.4	13.2	7.4
36	358	29.8	14.9	9.5
48	315	31.4	16.3	10.9
72	247	33.4	16.6	11.5
96	190	35.2	16.7	12.8
120	148	35.8	16.4	14.1
144	74	31.7	16.3	13.6
168	53	25.0	14.2	11.1
12–168	2223	30.3 [24.5, 35.1]	14.4 [12.4, 16.6]	9.4 [8.0, 10.5]

Table 2: Pooled (12–168 h) maximum-wind MAE (kt) with 95% storm-bootstrap confidence intervals, by subset.

Subset	n	Storms	AIFS raw	AIFS+GBM	NHC OFCL
All cases	2223	46	30.3 [24.5, 35.1]	14.4 [12.4, 16.6]	9.4 [8.0, 10.5]
RI cases only	162	13	71.4 [61.7, 77.5]	38.3 [32.0, 42.8]	20.2 [16.5, 23.5]
North Atlantic	772	15	32.4 [22.0, 39.3]	15.5 [10.7, 19.7]	10.4 [7.7, 12.4]
East Pacific	1451	31	29.2 [22.2, 35.7]	13.9 [11.4, 16.2]	8.9 [7.4, 10.2]
2025 (AIFS 1.0)	1526	30	29.6 [23.2, 35.5]	15.5 [12.9, 18.1]	9.6 [8.2, 10.9]
2026 (AIFS 2.0)	697	16	31.8 [20.1, 39.2]	12.1 [8.4, 14.5]	8.8 [5.8, 10.9]

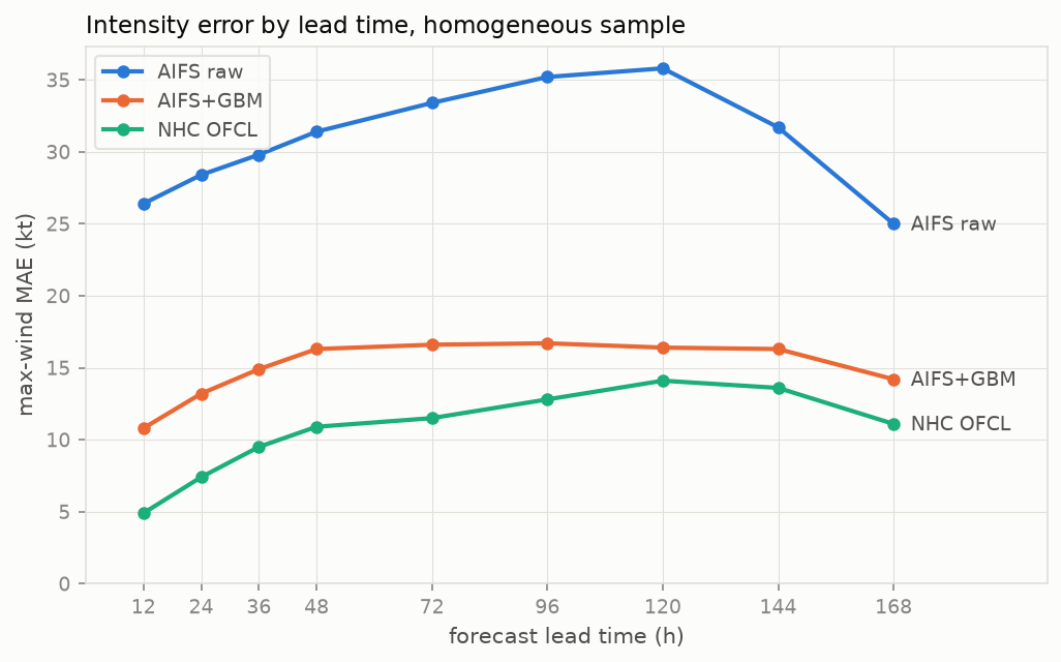


Figure 1: Maximum-wind MAE by lead time on the homogeneous sample. The raw AIFS error is dominated by the under-resolution bias; the GBM correction removes most of it at every lead.

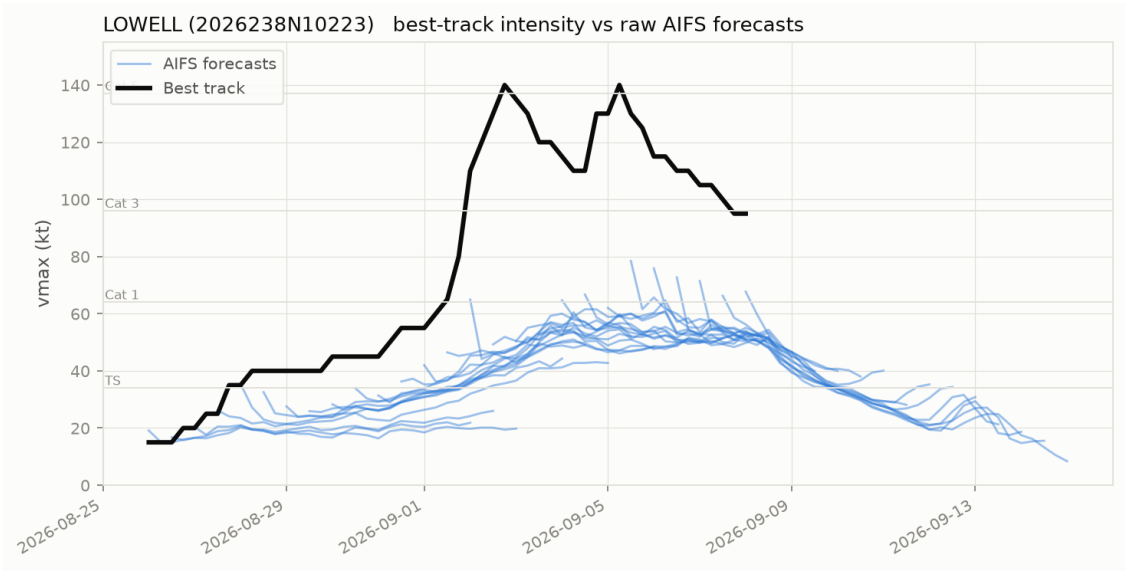


Figure 2: Observed intensity of Hurricane Lowell (2026, EP; black) against all 27 raw AIFS forecasts (blue). The model's track is good but its intensity saturates near 60 kt against an observed peak of 140 kt: the under-resolution bias the correction targets.

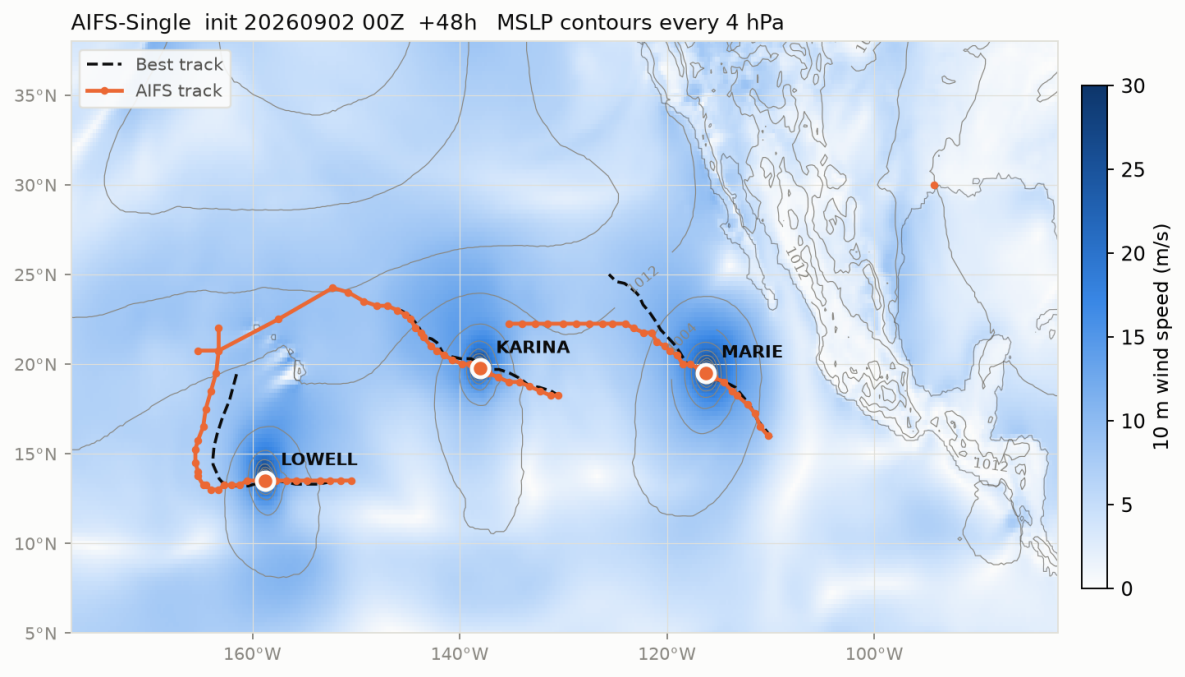


Figure 3: AIFS-Single forecast fields at +48 h from the 2026-09-02 00Z run: 10 m wind speed (shading), MSLP (contours every 4 hPa), forecast tracks (orange) and best tracks (dashed) for hurricanes Lowell, Karina and Marie.

5 Evaluation protocol for everything that follows

Three conventions hold for Sections 6 to 8. (i) Models predict the residual $r = v_{\text{obs}} - v_{\text{AIFS}}$ and are scored on the corrected wind $v_{\text{AIFS}} + \hat{r}$. (ii) Cross-validation is storm-grouped and five-fold; a temporal holdout (train on 2025 initializations, test on 2026: 20 storms, 4121 cases, 222 RI cases from 4 storms) is reported where the model is cheap enough to refit. (iii) Any difference between two models is a *paired* difference, with a 95% interval from resampling storms with replacement 2000 times. Treating cases as independent would understate the uncertainty by a factor of two or more: a quick check on twenty years of NHC forecasts found that nominal 95% intervals built on cases cover the truth 65 to 73% of the time, while the storm bootstrap covers 95% overall but only 86 to 93% on extreme subsets. Every number on an extreme subset therefore also states the number of storms behind it.

6 Eta-learning: a tail-matching objective

Method. Chang and Sapsis (2026) propose training a regressor with the usual mean-squared error plus a penalty that pulls the upper quantiles of its *predictions* toward the same quantiles of a reference distribution ν_0 of the target. Here the network is a 64-64 ReLU multilayer perceptron on the 22 features of Section 3, trained full-batch with Adam (learning rate 3×10^{-3}, 600 iterations, three seeds averaged), the penalty is the mean absolute gap between 25 quantiles (levels 0.95 to 0.999) of the corrected wind over the training fold and the same quantiles of the observed wind in that fold, and the weight λ is set every 100 iterations by the ratio of gradient norms of the two terms, as in the paper. The plain-MSE network with the penalty switched off ("ERM") is the natural control.

Ablation. Table 3 separates the three things that differ between the paper's GBM and the eta network: architecture, objective, and the 2× RI weighting.

The clean contrast is eta against ERM on the same network: -4.6 kt $[-6.8, -2.2]$. Against the GBM the overall difference is a tie (-0.7 kt $[-2.0, +0.8]$), but on RI cases eta wins by -10.5 kt $[-14.4, -5.6]$. Eta is also the only model whose forecasts reach the observed tail: its 0.95 and 0.99 forecast quantiles are 119 and 143 kt against 120 and 140 observed, whereas the GBM's are 98 and 118.

Is it calibration or climatology? The reference ν_0 above is the 18-month training sample itself, so the penalty can only be *calibrating* the network to a distribution it already sees. The version that imports information uses ν_0 from the IBTrACS archive since 1980 (1651 storms). The result barely moves: MAE 12.8 against 13.1, difference -0.3 kt $[-1.3, +0.8]$, and on RI $+2.3$ kt $[-1.4, +5.9]$. Eta is doing calibration, not importing climatology.

Lead time and the future season. Eta's advantage over ERM grows with lead (-3.7, -4.0, -4.7, -5.3, -5.8 kt across the bands 6–24, 30–48, 54–72, 78–120 and 126–168 h, every interval excluding zero), contrary to the expectation that matching a point forecast to the observed tail should hurt at long leads, where the best point forecast is less extreme than the observations. The reason is visible in the tail itself: at 126–168 h eta's 0.99 forecast quantile is 134 kt against 153 observed, so the penalty is no longer binding there. On the temporal holdout (train 2025, test 2026) the picture is unchanged: raw 27.8 kt, GBM 15.8, ERM 22.4, eta 15.6; eta minus GBM -0.2 kt $[-3.0, +3.5]$ overall and -7.9 kt $[-15.3, -0.5]$ on RI, the latter resting on four storms.

Table 3: Ablation grid: MAE (kt) on all 9149 cases and on the 585 RI cases (15 storms), storm-grouped 5-fold CV. "2×RI" doubles the weight of RI cases in training.

Architecture	Objective	All	RI
GBM	MSE	13.8	40.4
GBM	MSE, 2×RI	14.1	41.1
MLP	MSE (ERM)	17.7	36.6
MLP	MSE, 2×RI	16.6	33.4
MLP	eta	13.1	29.9
MLP	eta, 2×RI	13.5	26.1

Table 4: The fair baseline. MAE (kt), storm-grouped 5-fold CV, 3 seeds, 52 storms; tail quantiles of the forecasts against 120 / 140 observed. Early stopping selects 100 to 250 iterations, confirming that the 600-iteration network overfits.

Model	All	RI	$q_{0.95}$ / $q_{0.99}$
Plain MLP, 600 iterations	17.66	36.78	123 / 160
Tuned MLP (early stopping, weight decay)	14.42	35.45	112 / 133
Tuned MLP + quantile mapping to ν_0	15.44	34.71	122 / 160
Eta	13.29	29.88	118 / 148
GBM	14.12	41.12	98 / 118

6.1 The fair baseline: a tuned network plus quantile mapping

The ERM network above has no early stopping, and its tail overshoots (0.99 quantile 160 kt against 140 observed), so part of eta's gain could be regularization rather than the objective. The fair comparison, suggested by the advisor, is (a) the same network with early stopping and weight decay chosen on held-out training storms, and (b) that network followed by quantile mapping of its corrected winds onto ν_0, the classical marginal correction of climate science (Maraun, 2013). If eta still wins against (b), it is doing more than matching a histogram. All five models in Table 4 were trained in one run with identical folds and seeds.

Three of the 4.4 kt that eta gains over the untuned network are regularization: the tuned network alone closes the gap to 1.1 kt. That remaining 1.1 kt $[-2.3, -0.03]$ and the 5.6 kt on RI are the real effect of the objective. Quantile mapping, the post-hoc alternative, is *worse* than the tuned network by 1.0 kt overall: forcing a point forecast to have the observed marginal inflates its errors on ordinary cases, the well-known inflation problem of marginal correction. Eta beats the mapped baseline by 2.1 kt $[-3.2, -1.0]$, so it is not reducible to calibration after the fact. Figure 4 shows the same comparison by lead and Figure 5 the upper tail of each model.

Table 5: Paired MAE differences (kt) between the models of Table 4, with 95% storm-bootstrap intervals. Negative favours the first model.

Difference	All	RI (15 storms)
Eta − plain MLP	−4.36 [−7.09, −1.76]	−6.89 [−14.71, +2.08]
Eta − tuned MLP	−1.12 [−2.30, −0.03]	−5.57 [−9.33, −0.95]
Eta − tuned MLP + QM	−2.14 [−3.23, −0.99]	−4.83 [−8.14, −0.62]
Tuned MLP + QM − tuned MLP	+1.02 [+0.09, +2.06]	−0.73 [−1.90, +0.83]
Eta − GBM	−0.83 [−2.18, +0.51]	−11.24 [−15.26, −6.44]

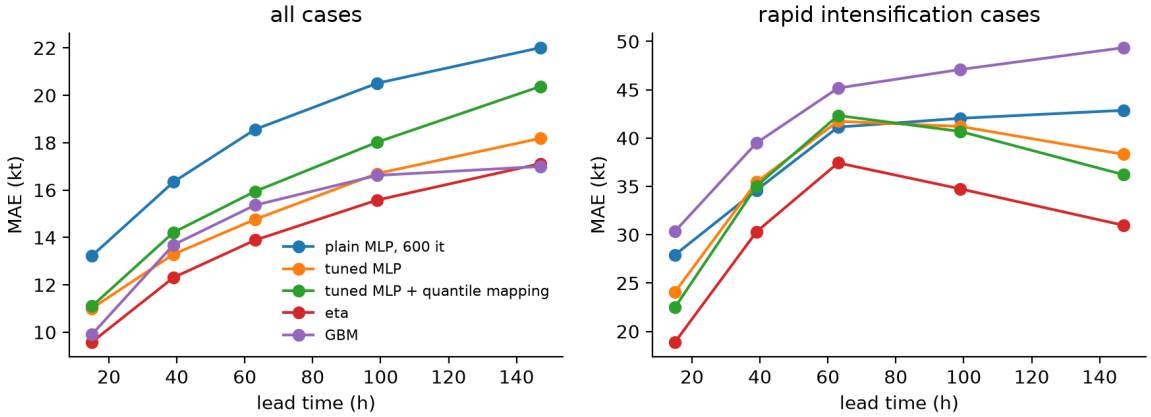


Figure 4: MAE by lead time for the five models of Table 4, all cases (left) and RI cases (right), out-of-fold. Eta's RI advantage is present at every lead.

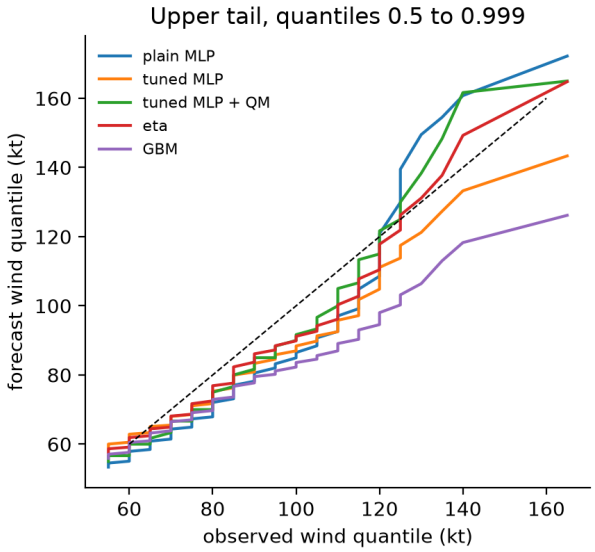


Figure 5: Forecast against observed wind quantiles from 0.5 to 0.999, out-of-fold. The GBM stops at 125 kt; the untuned network and the mapped network overshoot; eta follows the diagonal most closely.

6.2 Right tails, wrong storms

Matching the pooled tail does not mean the extreme forecasts land on the right storms. Among eta's 440 forecasts of 120 kt or more, the observed wind is on average 22 kt lower ($[-33.5, +1.7]$), and 358 of the 440 come from two storms (Lowell and Erin). Of the 636 cases observed at 120 kt or more, eta forecasts 120 kt or more for 26%. This is not specific to eta: the untuned network overshoots by 34 kt on its own extreme forecasts, and the GBM followed by quantile mapping shows the same pattern (-23 kt). It is the generic behaviour of a point forecast whose marginal has been forced onto the observed one, and the eta paper states the limitation itself ("distributionally calibrated candidate extremes, rather than ... true location, time, or mechanism"). An attempt to fix it by applying the penalty within groups (by initial intensity) was abandoned once the group-wise gaps were bootstrapped: every interval includes zero. The constructive response is to stop forcing a point forecast and predict a distribution instead (Section 8).

7 Environmental predictors from the forecast itself

The 22 features so far describe the AIFS vortex (wind, pressure, their trends) and the storm's history, nothing about its environment. The operational SHIPS scheme (DeMaria and Kaplan, 1994; DeMaria et al., 2005) and its RI index (Kaplan and DeMaria, 2003) rest on a handful of environmental quantities: vertical wind shear, mid-level humidity, upper-level divergence and temperature, low-level vorticity, and the ocean's potential intensity. All of them can be computed from the AIFS forecast's own pressure-level fields, which makes them legitimate forecast-time features. From the fields of Section 2, each case receives: the magnitude and direction of the ring-mean (200 to 800 km from the forecast centre) 200 minus 850 hPa wind vector; relative humidity at 700 and 500 hPa from q and T; 200 hPa divergence and 850 hPa relative vorticity within 1000 km; 200 hPa temperature; skin temperature within 200 km as a sea-surface proxy; the DeMaria and Kaplan (1994) maximum potential intensity from it; and the potential minus the current forecast wind. Each is taken at the forecast lead, at initialization, and averaged along the forecast track, for 36 features in all.

Table 6 gives the effect on the point GBM. The gain is about 1 kt, concentrated at short leads, with intervals that touch zero, and nothing on RI. The advisor's hypothesis was that the information bottleneck, not the objective, limits RI skill; these predictors do not lift it, which leaves the AIFS environment at 0.25°, the single deterministic run, or the sample of 15 RI storms as the remaining suspects.

Table 6: Point GBM with and without the 36 environmental predictors. MAE (kt), storm-grouped 5-fold CV, paired storm-bootstrap intervals.

Subset	n	Storms	Base	+ SHIPS	Difference
All cases	9149	52	13.99	13.08	-0.91 $[-2.13, +0.17]$
RI cases	585	15	40.62	39.60	-1.02 $[-6.39, +3.67]$
Lead ≤ 48 h	4259	52	11.56	10.50	-1.06 $[-2.29, 0.00]$
Lead > 96 h	2238	29	16.70	16.81	$+0.11$ $[-1.93, +1.84]$

8 A probabilistic correction

Method. Nine gradient-boosted regressors with the pinball loss predict the residual quantiles at levels 0.02, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95 and 0.98 (same features as Table 6 with SHIPS, same tree hyperparameters and RI weighting as the point GBM, quantiles sorted to remove crossings). Quantile trees turned out to be far too confident on unseen storms: the raw 10–90% interval covered only 51% of out-of-fold cases. The remedy is a cross-conformal recalibration in the spirit of Romano et al. (2019) and Barber et al. (2021): within each training fold, an inner storm-grouped 5-fold produces out-of-fold residuals $y - \hat{q}_\tau(x)$ for every level τ; the empirical τ-quantile of those residuals, computed separately in three lead bands (0–48, 54–96, 102–168 h), is added to $\hat{q}_\tau$ at test time. Scores are the mean absolute error of the median, the continuous ranked probability score (CRPS, approximated by twice the mean pinball loss over the nine levels (Gneiting and Raftery, 2007)), and the coverage of the 10–90% interval. The probability of RI is read off the predicted distribution as $P(v_t \geq v_{t-24} + 30\,\text{kt})$ and scored with the Brier score against climatology; a direct gradient-boosted classifier on the RI label is the baseline.

Calibration. Figure 6 shows the fraction of observations falling below each predicted quantile. The raw quantile trees are badly over-confident (10–90% interval covering 51%); after recalibration the curve sits on the diagonal in cross-validation (82%) and on the future season (84%), and the CRPS improves from 7.69 to 6.69 kt. Calibration is not uniform across cases, however: on RI cases the 80% interval covers only 35% in CV, and on cases observed at 96 kt or more it covers 51%. The price of covering ordinary cases honestly while being unable to tell which storms will intensify is a wide upper tail: the 0.99 quantile of the predicted 95th percentile is 170 kt in CV against 140 observed. The model knows it does not know.

Accuracy. Table 7 compares the median of the recalibrated distribution with the point GBM trained on the same features. In cross-validation the median is better by 0.85 kt $[-1.57, -0.15]$ overall and by 2.4 kt $[-4.7, -0.2]$ on cases observed at 96 kt or more; on RI cases the difference (-2.2 kt) has an interval that includes zero. On the 2026 holdout the overall difference reverses ($+0.4$ kt $[-0.1, +1.2]$) because the median is worse at leads of 48 h or less, while it stays better at long leads and on strong storms. The honest summary is that the probabilistic model costs nothing in point accuracy and adds calibrated intervals.

Probability of rapid intensification. This is where the project set out to arrive, and the answer is negative. Read off the predicted distribution, $P(\text{RI})$ has a Brier skill score of -0.06 against climatology in cross-validation and -0.44 on the holdout, and the reliability diagram (Figure 7) shows why: when the model says 0.6, RI follows 21% of the time in CV and 5% on the holdout. The direct classifier on the RI label does no better in CV (-0.03 with SHIPS features, -0.05 without),

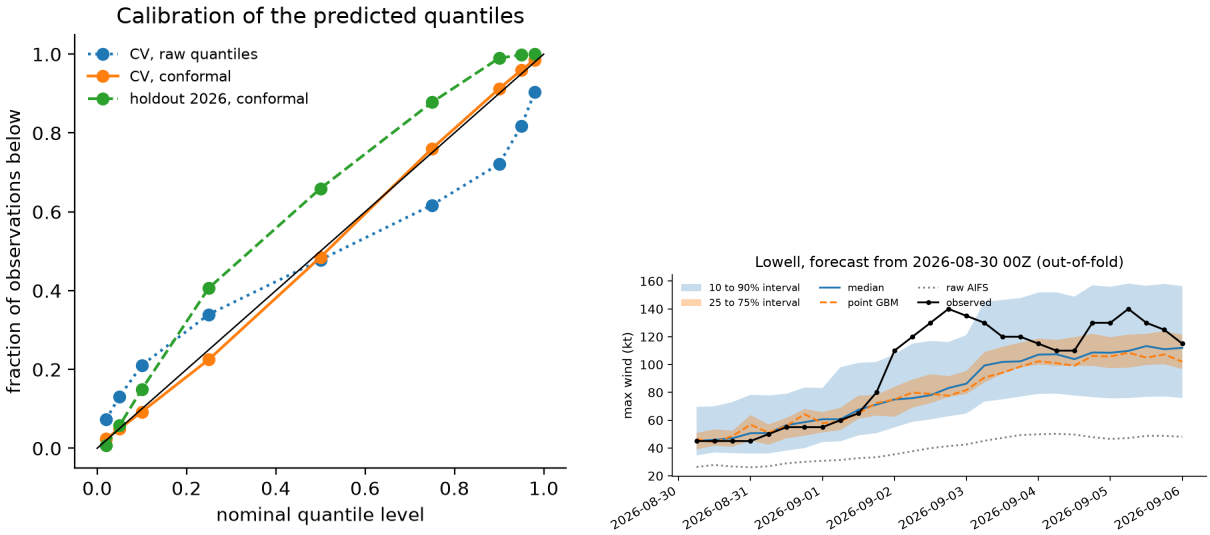


Figure 6: Left: calibration of the predicted quantiles before and after the cross-conformal step, in cross-validation and on the 2026 holdout. Right: an out-of-fold forecast of Hurricane Lowell (2026): raw AIFS saturates near 50 kt, the corrected median follows the storm to about 110 kt, and the observed rapid intensification to 140 kt leaves even the 10–90% band.

and on the holdout its score flips sign with the feature set (+0.08 without SHIPS, −0.48 with), which with 222 RI cases from four storms is noise. Neither the objective nor the environmental predictors turn 15 RI storms into a usable RI probability. The eta network's RI advantage of Section 6.1 is a gain in the *magnitude* of its forecasts on storms that did intensify, not an ability to say in advance which storms will.

Table 7: Median of the recalibrated quantile GBM against the point GBM, same features (base plus SHIPS). MAE (kt), paired storm-bootstrap intervals, CRPS and 10–90% interval coverage of the quantile model.

Evaluation	Subset	n	Storms	Median	Point	Difference	CRPS	Coverage
CV	all	9149	52	12.78	13.63	-0.85 $[-1.57, -0.15]$	6.69	82%
CV	RI	585	15	40.66	42.82	-2.16 $[-5.69, +0.72]$	18.10	35%
CV	lead ≤ 48 h	4259	52	10.28	10.80	-0.52 $[-1.13, +0.04]$	5.52	82%
CV	lead > 96 h	2238	29	16.36	17.81	-1.45 $[-2.82, +0.08]$	8.31	81%
CV	obs ≥ 96 kt	1444	13	33.10	35.49	-2.39 $[-4.70, -0.19]$	14.99	51%
Holdout	all	4121	20	14.07	13.64	$+0.43$ $[-0.14, +1.24]$	7.49	84%
Holdout	RI	222	4	32.81	35.77	-2.95 $[-5.72, +0.83]$	12.03	86%
Holdout	lead ≤ 48 h	1800	20	13.58	12.52	$+1.06$ $[+0.54, +1.65]$	6.98	78%
Holdout	lead > 96 h	1117	12	14.16	15.26	-1.10 $[-2.69, +0.08]$	7.88	93%

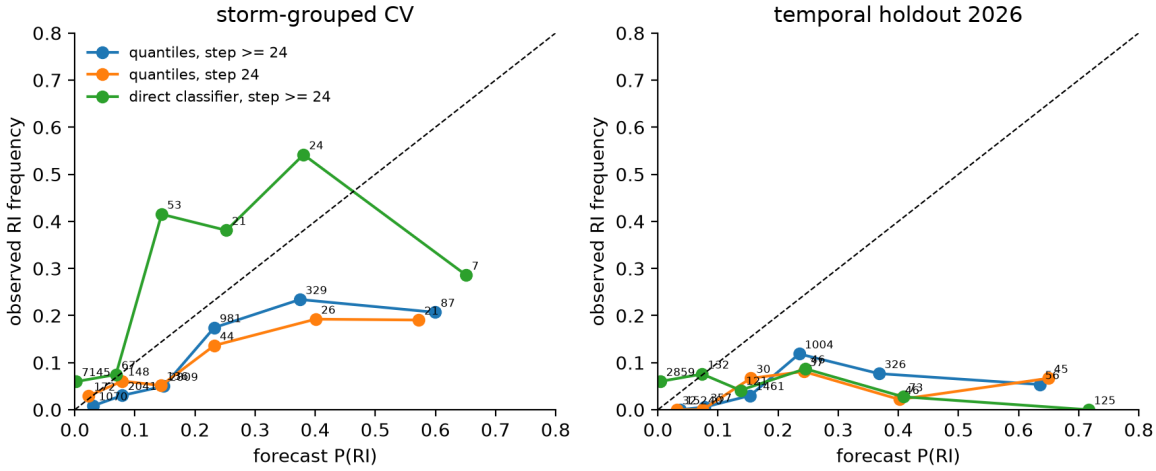


Figure 7: Reliability of the RI probability from the predicted distribution and from a direct classifier, in cross-validation (left) and on the 2026 holdout (right); labels give the number of cases in each bin. A skilful forecast would follow the diagonal.

9 What did not work

Two further ideas were tested and dropped; both are recorded so that they are not retried. *Wind through pressure.* The observed wind-pressure relationship is tight (6.9 kt MAE from observed pressure), so correcting AIFS pressure and converting might beat correcting wind directly. It does not: 15.2 against 14.0 kt, worse by $+1.2$ kt $[+0.1, +2.3]$, because the correction removes 51% of the wind error but only 13% of the pressure error. *An archive prior.* A model of future intensity from storm history alone, trained on 45 years of IBTrACS with no forecast input, reaches 17.6 kt on these storms and beats the AIFS-based correction at 6–24 h (7.2 against 9.9). Stacked as a feature it helps when training storms are few (-2.0 kt $[-2.8, -1.1]$ with 10 storms) and not with all 41 (-0.3 $[-0.7, +0.2]$); a Bayesian combination of the archive prior with an AIFS likelihood beat stacking only at 5 training storms and was worse on RI. With a full season of paired data the archive adds nothing.

Table 8: Pooled maximum-wind MAE (kt), this work versus AIFS-TC (their NA+EP test set, 29 storms). Samples differ; compare each system to its own sample's OFCL.

	This work	AIFS-TC
AIFS raw	30.3	35.3
Corrected	14.4	11.0
NHC OFCL	9.4	10.7
AIFS raw, RI only	71.4	71.1
Corrected, RI only	38.3	20.9
OFCL, RI only	20.2	20.1

10 Comparison with AIFS-TC

Table 8 sets these results against the paper's NA+EP evaluation (Allen et al., 2026). The samples differ (their held-out 2025 under one model version and hindcast initial conditions; here 2025–2026 operational output under two versions), so each column should be read against its own OFCL, not across columns.

Three observations. First, the *baselines* land on the paper's numbers: raw AIFS error is the same phenomenon at the same magnitude, and on RI cases the agreement is striking (71.4 vs. 71.1 kt). Together with the 1.1 kt tracker agreement, this indicates the pipeline measures the same quantity the paper measures. Second, the *correction* reproduces the qualitative claim, recovering roughly three quarters of the raw-to-OFCL gap, but not the paper's parity with OFCL. Third, the shortfall concentrates exactly where limited training data should hurt most: rare RI events, of which this dataset contains 13 storms' worth against the paper's nine years. The 2026-only result (12.1 kt, single model version) already approaches the paper's 11.0.

11 Limitations

One and a half seasons versus nine years of training data, and 52 storms of which 15 produced RI and 9 reached 120 kt: every RI or tail statistic in this report rests on a handful of storms, and the intervals say so. Two model versions in one dataset, handled only by a feature flag. The GBM branch of AIFS-TC only (no CNN over three-dimensional patches), and a single deterministic AIFS run: the AIFS ensemble is in the same archive from July 2025 and its spread was not used. Best-track initial intensity as a feature where fully operational practice would use the real-time CARQ estimate (the paper reports both variants); two basins; leads capped at 168 h. Position errors are measured but not corrected. The eta network was tuned once (the paper's settings) and the quantile model inherits the point GBM's hyperparameters; neither was searched. Model variants for the dropped ideas of Section 6.2 and the archive prior were chosen after seeing results and are reported as negative findings only.

12 Reproducibility

All code is at `https://github.com/MichelFaloughi/scmp-5990`: the data pipeline in `pipeline/` (catalog, fetch, track, correct, evaluate, and the pressure-level fetch and SHIPS predictor scripts), the modeling of Sections 6 to 8 in `core/` (`gbm.py`, `eta.ipynb`, `paired_ci.ipynb`, `fair_baseline.py`, `quantile.py`, `figures.py`). The 7 GB of surface fields and 23 GB of pressure-level fields regenerate

from public sources with the fetch scripts (a few hours over the network); the modeling runs on a laptop in under an hour. Data sources: ECMWF Open Data (CC-BY-4.0), IBTrACS v04r01, and the NHC ATCF archive.

References

Anna Allen, Wessel P. Bruinsma, Michael Maier-Gerber, Harrison Cook, Matthew Chantry, and Richard E. Turner. AIFS-TC: A simple correction competitive with the operational frontier for tropical cyclone intensity forecasting. *arXiv preprint arXiv:2608.09959*, 2026.

Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Predictive inference with the jackknife+. *The Annals of Statistics*, 49(1):486–507, 2021.

Shih-Chun Chang and Themistoklis P. Sapsis. Eta-learning: learning to forecast rare and extreme events by matching tail statistics. *arXiv preprint arXiv:2510.19161*, 2026. Accepted, Nature Communications.

Mark DeMaria and John Kaplan. A statistical hurricane intensity prediction scheme (SHIPS) for the Atlantic basin. *Weather and Forecasting*, 9(2):209–220, 1994.

Mark DeMaria, Michelle Mainelli, Lynn K. Shay, John A. Knaff, and John Kaplan. Further improvements to the statistical hurricane intensity prediction scheme (SHIPS). *Weather and Forecasting*, 20(4):531–543, 2005.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

Google DeepMind. Weather Lab: experimental AI tropical cyclone predictions. `https://deepmind.google/science/weatherlab`, 2025. Ensemble mean appears in ATCF a-decks as GDMN.

John Kaplan and Mark DeMaria. Large-scale characteristics of rapidly intensifying tropical cyclones in the North Atlantic basin. *Weather and Forecasting*, 18(6):1093–1108, 2003.

Kenneth R. Knapp, Michael C. Kruk, David H. Levinson, Howard J. Diamond, and Charles J. Neumann. The international best track archive for climate stewardship (IBTrACS): Unifying tropical cyclone data. *Bulletin of the American Meteorological Society*, 91(3):363–376, 2010.

Simon Lang, Mihai Alexe, Matthew Chantry, Jesper Dramsch, Florian Pinault, Baudouin Raoult, Mariana C. A. Clare, Christian Lessig, Michael Maier-Gerber, Linus Magnusson, Zied Ben Bouallègue, Ana Prieto Nemesio, Peter D. Dueben, Andrew Brown, Florian Pappenberger, and Florence Rabier. AIFS: ECMWF's data-driven forecasting system. *arXiv preprint arXiv:2406.01465*, 2024.

Douglas Maraun. Bias correction, quantile mapping, and downscaling: revisiting the inflation issue. *Journal of Climate*, 26(6):2137–2143, 2013.

Yaniv Romano, Evan Patterson, and Emmanuel Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems*, volume 32, 2019.